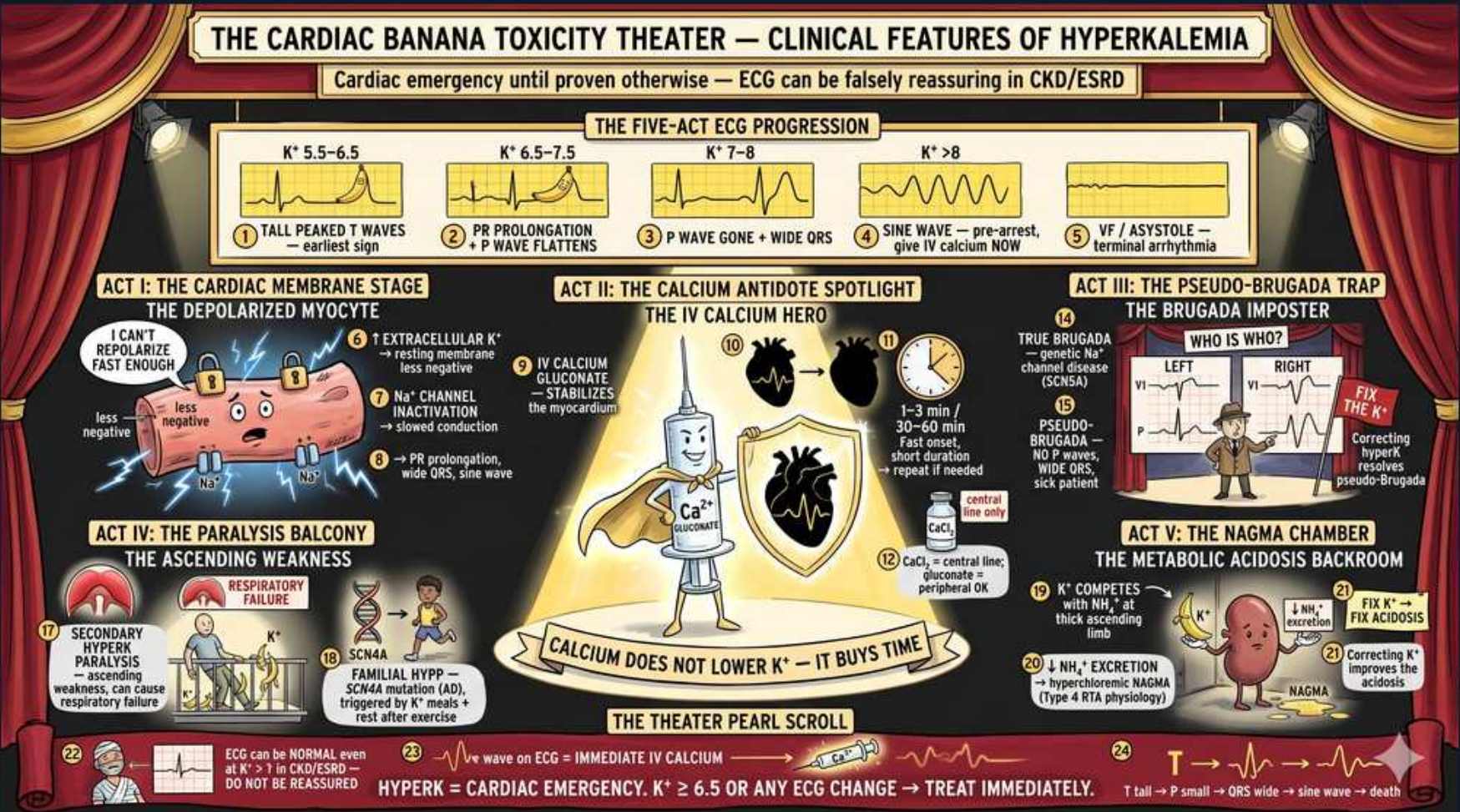


Hyperkalemia — clinical features & ECG



1. Cardiac emergency

WHAT YOU SEE

The theater-marquee banner 'CARDIAC EMERGENCY UNTIL PROVEN OTHERWISE — ECG can be falsely reassuring in CKD/ESRD' over the stage — sets the whole scene as an emergency driven by the ECG, not the number.

WHAT IT ENCODES

Hyperkalemia is a cardiac emergency because elevated extracellular K⁺ destabilizes the cardiac resting membrane potential, risking lethal arrhythmia. The ECG — not the absolute number — drives urgency: any ECG change or K⁺ ≥ 6.5 mandates immediate IV calcium. Disposition hinges on rate of rise and ECG, so get a STAT ECG and place the patient on a monitor first.

BOARD TRAP

WRONG to wait for a confirmatory repeat K⁺ before acting when the ECG already shows peaked T waves or wider changes — and equally WRONG to be reassured by a NORMAL ECG in CKD/ESRD, where the myocardium is chronically adapted and the ECG can stay normal even at K⁺ > 7. The discriminator: treat based on ECG plus clinical context, never on a single number in isolation.

2. Peaked T (5.5-6.5)

WHAT YOU SEE

Act-1 ECG strip showing tall, narrow, tented PEAKED T waves (K⁺ 5.5-6.5) — the spiky T is the earliest curtain-raiser change.

WHAT IT ENCODES

The earliest ECG change is tall, peaked/tented, narrow-based T waves, typically appearing at K⁺ 5.5-6.5 mM. They reflect accelerated repolarization from increased K⁺ conductance. This is the cue to start treatment even before higher-grade changes appear.

BOARD TRAP

WRONG to call any tall T wave hyperkalemia — hyperacute T waves of early STEMI are broad-based and asymmetric, whereas hyperkalemic T waves are narrow, symmetric, and tented. Discriminator: check the K⁺ and the base of the T wave; STEMI T waves come with reciprocal ST changes and a clinical ischemic story.

3. PR prolong / flat P (6.5-7.5)

WHAT YOU SEE

Act-2 ECG strip with a stretched PR interval and a flattening P wave (K⁺ 6.5-7.5) — the fading P pictures slowing atrial/AV conduction.

WHAT IT ENCODES

At K⁺ ~6.5-7.5 mM the PR interval prolongs and the P wave flattens as atrial myocardium and AV conduction slow. This marks progression from 'early' to 'moderate-severe' hyperkalemia and increased urgency. IV calcium is indicated at this stage.

BOARD TRAP

WRONG to interpret PR prolongation here as primary first-degree AV block needing only observation. Discriminator: in the setting of a high K⁺, conduction slowing is the metabolic emergency — fix the potassium, and stabilize with calcium; do not simply 'monitor the block.'

4. Wide QRS (7-8)

WHAT YOU SEE

Act-3 ECG strip where the P wave has vanished and the QRS is fat/widened (K⁺ 7-8) — the broad complex is the pre-arrhythmic warning.

WHAT IT ENCODES

At K⁺ ~7-8 mM the P wave disappears entirely (sinoventricular rhythm) and the QRS widens from progressive Na⁺ channel inactivation and slowed intraventricular conduction. A wide QRS in this setting is pre-arrhythmic — VF/asystole may be imminent. Give IV calcium immediately and prepare for repeat dosing.

BOARD TRAP

WRONG to treat a wide-complex rhythm here as a primary bundle branch block or VT and reach for antiarrhythmics. Discriminator: absent P waves + widening QRS + known/likely hyperkalemia = give calcium first; lidocaine/amiodarone will not fix the membrane and can worsen the situation.

5. Sine wave (>8)

WHAT YOU SEE

Act-4 ECG strip melted into a smooth SINE WAVE (K⁺ >8) — QRS and T fused into a rolling wave, the pre-terminal rhythm.

WHAT IT ENCODES

At K⁺ > 8 mM the widened QRS merges with the T wave into a sinusoidal 'sine wave' pattern — the pre-terminal rhythm. This demands IV calcium NOW (calcium gluconate 1-2 g, or chloride via central line) plus simultaneous K⁺-lowering therapy. Without immediate calcium, this degenerates to VF or asystole within minutes.

BOARD TRAP

WRONG to prioritize insulin/dextrose or dialysis access before giving calcium when a sine wave is present. Discriminator: calcium is the only agent that acts in 1-3 min to stabilize the membrane; shift and removal therapies are too slow to prevent imminent arrest.

6. VF / asystole

WHAT YOU SEE

The final ECG strip collapsing into chaotic VF then a flat ASYSTOLE line — the curtain-call terminal rhythm.

WHAT IT ENCODES

The terminal rhythm of untreated hyperkalemia is ventricular fibrillation or asystole. In cardiac arrest with suspected/known hyperkalemia, give empiric IV calcium plus high-dose shift therapy during CPR. Hyperkalemia is one of the reversible 'H's of arrest and must be treated empirically.

BOARD TRAP

WRONG to run a standard ACLS arrest algorithm without empiric calcium when hyperkalemia is suspected (dialysis patient, crush injury, missed dialysis). Discriminator: in arrest, do not wait for a lab value — give IV calcium and treat the reversible cause concurrently with standard ACLS.

7. Resting membrane up

WHAT YOU SEE

Act 1's DEPOLARIZED MYOCYTE cartoon drawn 'less negative' inside — the raised resting potential creeping toward threshold.

WHAT IT ENCODES

High extracellular K⁺ reduces the K⁺ gradient across the membrane, raising (making less negative) the resting membrane potential toward threshold per the Nernst equation. Initially this makes cells MORE excitable, but sustained depolarization then inactivates voltage-gated Na⁺ channels. This biphasic effect explains both early excitability and later conduction failure.

BOARD TRAP

WRONG to assume that a 'less negative' resting potential means the heart simply fires more easily and is therefore more robust. Discriminator: persistent partial depolarization holds Na⁺ channels in the inactivated state, so net conduction SLOWS — the dominant danger is failed conduction, not hyperexcitability.

8. Na channel inactivation

WHAT YOU SEE

The Na⁺ channel cartoon stuck in its INACTIVATED shape with a 'slowed conduction' arrow — jammed channels picture the failing phase-0 upstroke.

WHAT IT ENCODES

Sustained membrane depolarization locks voltage-gated Na⁺ channels in their inactivated state, reducing phase-0 upstroke velocity and slowing conduction. This is the mechanistic basis for PR prolongation, P-wave loss, and QRS widening. IV calcium works by raising the threshold potential, restoring the gap between resting and threshold and re-enabling Na⁺ channel availability.

BOARD TRAP

WRONG to think calcium lowers serum K⁺ to fix conduction. Discriminator: calcium acts purely electrophysiologically — it shifts threshold to restore excitability/conduction within minutes, while serum K⁺ stays elevated until shift/removal therapy works.

9. IV calcium stabilizes

WHAT YOU SEE

Act 2's caped IV-CALCIUM HERO wielding a Ca²⁺ shield/syringe — the bodyguard stabilizes the membrane (raises threshold) without lowering K⁺.

WHAT IT ENCODES

IV calcium is the first drug given in any hyperkalemic emergency: calcium gluconate 1-2 g (10 mL of 10%) IV, or calcium chloride via central access (3x the elemental calcium, more sclerosing). It antagonizes the membrane effects of K⁺ by raising threshold potential, restoring the resting-to-threshold gap. Onset is 1-3 minutes; redose if ECG changes persist after 5 minutes.

BOARD TRAP

WRONG ANSWER on boards: 'calcium gluconate lowers potassium.' Calcium does NOT lower K⁺ at all — it only stabilizes the myocardium (buys time). Discriminator: pair calcium (membrane protection) with insulin+glucose / albuterol (shift) and dialysis / kayexalate-patiromer / loop diuretics (removal) — these three roles are distinct and tested.

10. Onset 1-3 min

WHAT YOU SEE

The hourglass/clock beside the calcium hero reading 1-3 min onset, ~30-60 min duration ('repeat if needed') — the short timer pictures why calcium only buys time.

WHAT IT ENCODES

Calcium's protective effect begins in 1-3 minutes and lasts only ~30-60 minutes, so it must be repeated if ECG changes recur and bridged with definitive K⁺-lowering therapy. Because it is transient, calcium alone never constitutes complete treatment. Always co-administer shift agents (insulin/glucose, albuterol) and arrange removal (dialysis, GI binders, diuresis).

BOARD TRAP

WRONG to give calcium, see the ECG improve, and then stop active management. Discriminator: the short duration means K⁺ will re-injure the membrane once calcium wears off unless shift and removal are underway — calcium buys a 30-60 minute window, not a cure.

11. Pseudo-Brugada

WHAT YOU SEE

Act 3's BRUGADA IMPOSTER — a masked figure faking the coved V1-V2 ST elevation ('WHO IS WHO?') — the disguise pictures hyperkalemia mimicking Brugada, which resolves on correcting K⁺.

WHAT IT ENCODES

Severe hyperkalemia can produce a coved ST-elevation pattern in V1-V2 that mimics Brugada — termed pseudo-Brugada or a Brugada phenocopy. Correcting the potassium fully resolves the ECG pattern. Recognizing this prevents misdiagnosis as the inherited channelopathy.

BOARD TRAP

WRONG ANSWER: diagnosing true Brugada syndrome and referring for an ICD based on this ECG. Discriminator: true Brugada is a genetic SCN5A Na-channel disease that persists regardless of electrolytes; pseudo-Brugada normalizes once K⁺ is corrected — always check and treat the K⁺ first.

12. Ascending paralysis

WHAT YOU SEE

Act 4's PARALYSIS BALCONY with a figure weak from the legs up and a respiratory-failure cue — the ascending weakness climbing the balcony pictures secondary hyperkalemic paralysis.

WHAT IT ENCODES

Beyond the heart, hyperkalemia can cause secondary hyperkalemic paralysis: ascending muscle weakness from sustained depolarization blocking skeletal Na⁺ channels. It typically begins in the legs and ascends, sparing cranial nerves and sensation. Weakness can progress to flaccid paralysis and respiratory compromise.

BOARD TRAP

WRONG to anchor on Guillain-Barre for an ascending paralysis without checking the potassium. Discriminator: GBS shows areflexia with albuminocytologic dissociation on LP, while hyperkalemic paralysis has a high serum K⁺ that resolves with treatment — always send electrolytes in acute ascending weakness.

13. Familial HyperKPP

WHAT YOU SEE

The SCN4A familial-HyperKPP cartoon: episodic paralysis triggered by rest-after-exercise and K⁺-rich meals — the skeletal Na-channel mutation drawn as the family curse.

WHAT IT ENCODES

Familial hyperkalemic periodic paralysis is an autosomal dominant SCN4A skeletal Na-channel mutation causing episodic weakness triggered by rest after exercise, potassium-rich meals, or cold. Attacks are short (minutes to hours) and may be aborted with carbohydrate/mild exercise. Serum K⁺ rises during attacks but may be only mildly elevated or normal.

BOARD TRAP

WRONG to confuse this with HYPOkalemic periodic paralysis (CACNA1S/SCN4A), which is triggered by carbohydrate loads and insulin and is treated by giving potassium. Discriminator: in HyperKPP attacks K⁺ rises and carbohydrates help; in HypoKPP attacks K⁺ falls and carbohydrates trigger — the trigger and direction of K⁺ change distinguish them.

14. NAGMA backroom

WHAT YOU SEE

Act 5's METABOLIC ACIDOSIS BACKROOM where K^+ competes with H^+ for distal secretion — the tug-of-war pictures Type-4-RTA normal-anion-gap acidosis.

WHAT IT ENCODES

Hyperkalemia and acidosis are linked: in the distal tubule, K^+ secretion competes with H^+ secretion, so high K^+ impairs acid excretion and can produce a normal-anion-gap metabolic acidosis (classically Type 4 RTA). Hyperkalemia also suppresses renal ammoniagenesis, blunting acid excretion further. This bidirectional relationship is high-yield.

BOARD TRAP

WRONG to assume metabolic acidosis with hyperkalemia is always a high-anion-gap process like DKA. Discriminator: hyperkalemia-associated Type 4 RTA is a NORMAL-anion-gap (hyperchloremic) acidosis from hypoaldosteronism — calculate the anion gap and check aldosterone status to separate it from gap acidoses.

15. Treat-now threshold

WHAT YOU SEE

The bottom THEATER PEARL banner: $K^+ \geq 6.5$ OR any ECG change → treat immediately (Calcium / shift / remove) — the action line spelled across the stage floor.

WHAT IT ENCODES

Definitive treat-now triggers: serum $K^+ \geq 6.5$ mM OR ANY ECG change at any K^+ level. Treatment proceeds in three parallel arms — stabilize (IV calcium), shift (insulin 10 units + 25 g dextrose, albuterol 10-20 mg neb), and remove (dialysis for refractory/ESRD, loop diuretics if making urine, GI binders patiromer/sodium zirconium). Mnemonic 'C BIG K Drop': Calcium, Bicarb (if acidotic), Insulin/Glucose, Kayexalate/binders, Diuretics/Dialysis.

BOARD TRAP

WRONG to defer treatment until the patient becomes symptomatic or until a higher K^+ returns. Discriminator: do not wait for symptoms — $K^+ \geq 6.5$ OR any ECG change is the action line; symptoms (weakness, arrhythmia) are late and may be the arrest itself.
